RUNGTA INTERNATIONAL SKILLS UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY
BACHELOR OF COMPUTER APPLICATION

PS No. 67

Project Title Hotel Room Booking System

Problem Statement

Build a console application for a hotel where the user can view available rooms, book a room for a guest for a number of nights, cancel a booking and generate the bill. Booking details are saved in a file.

Language: Python 3 **Concepts:** Dictionaries, Lists, Functions, File Handling

Note: Rooms (per night): 101 Single Rs. 2000, 102 Double Rs. 3500, 201 Suite Rs. 6000. Bill = rate x nights + 12% GST. Nights must be 1 to 30. Bookings are saved in bookings.txt.

Test Cases / Case Statements (4 cases)

INPUT	EXPECTED OUTPUT
Test Case 1: Normal Input	
Menu option 2 (Book) Guest: Amit Room: 101 Nights: 2	Room 101 booked for Amit | 2 nights Bill: 4000 + GST 480 = Rs. 4480
Test Case 2: Boundary / Minimum Input	
Menu option 2 (Book) Guest: Neha Room: 102 Nights: 1 (minimum)	Room 102 booked for Neha | 1 night Bill: 3500 + GST 420 = Rs. 3920
Test Case 3: Invalid Input	
Room 101 is already booked by Amit Menu option 2 (Book) Guest: Neha, Room: 101, Nights: 3	Room 101 is already booked. Please choose another room.
Test Case 4: Different Valid Scenario	
bookings.txt has: Room 101 - Amit - 2 nights (a) Menu option 1 (Available rooms) (b) Menu option 4 (Cancel): Room 101	(a) 102 Double Rs. 3500 | 201 Suite Rs. 6000 (b) Booking for room 101 cancelled. Room 101 is now available.

Bare Minimum Requirements

1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Store room details in a dictionary and current bookings in a list of dictionaries.
3. Provide view-available, book, cancel and bill generation through functions.
4. Validate room number and number of nights; do not allow double booking of a room.
5. Save bookings in bookings.txt after every change and load them at program start.

Instruction

1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.